AV Tech System Documentation

Mackie DL32SE Mixer and Master Fader SE Guide

Draft module for the church AV / streaming manual - prepared 2026-05-19; Figure 1 and audio-path notes revised 2026-06-23

Document status
This module is written without current Master Fader screenshots. It is suitable as a text/reference baseline now, and screenshots can be inserted later in the same style as the Task Scheduler and ER605 documentation modules.

1. Purpose of this document

This document explains the Mackie DL32SE digital mixer and the Master Fader SE control app as part of the church AV technology system. It is intended for a future volunteer, technician, or maintainer who needs to understand what the mixer does, why it was installed, how it fits the livestream system, and how auxiliary mixes are used for streaming and other speaker zones.

2. What the DL32SE is

The Mackie DL32SE is a rack-mount digital mixer. Unlike an older analog mixer with physical knobs and faders on the surface, the DL32SE is controlled primarily through software. It lives in the AV cabinet and provides microphone preamps, signal processing, routing, USB audio, and multiple independent output mixes.

Feature	Why it matters in this AV system
32 input channels with Onyx+ preamps	Enough inputs for wired microphones, wireless microphones, instruments, playback, and future expansion.
10 fully assignable XLR outputs	Allows main room sound, stream feeds, auxiliary speaker feeds, and monitor-style outputs to be routed separately.
8 stereo-linkable aux sends	Allows independent mixes for streaming, monitors, overflow rooms, or auxiliary amplifiers.
Master Fader SE control app	Allows control from the BeelinkOBS PC, tablet, or other approved devices on the AV network.
32x32 USB audio interface	Can send mixer audio to a computer and receive computer playback when configured. In the current Stream Agent/OBS setup, USB audio is not used.
Snapshots, presets, mute groups, VCAs, processing	Allows repeatable setup and faster recovery if settings are accidentally changed.

Mackie documentation describes the DL32SE as a 32-channel digital rack mixer controlled with Master Fader SE, with 32 inputs, 10 assignable XLR outputs, 8 stereo-linkable aux sends, and a 32x32 USB audio interface. See the source list at the end of this document.

3. What Master Fader SE is

Master Fader SE is the control app for the DL32SE. In normal use, the mixer hardware handles the audio, while Master Fader SE provides the control surface on a Windows PC, tablet, or other supported device. In this church system the app is part of the operator workflow because the physical mixer in the cabinet does not provide a traditional full-size fader surface.

- Use Master Fader SE to control channel faders, mutes, gain-related settings, EQ, dynamics, aux sends, outputs, routing, snapshots, and user access limits.
- The mixer connects by Ethernet to the church AV network. Wireless control from a tablet is provided by the AV access point, not by relying on a built-in Wi-Fi mixer hotspot.
- The BeelinkOBS PC can run Master Fader SE so the main streaming computer can also monitor or adjust the audio mix when required.
- Access limiting should be considered if other volunteers or tablets are allowed to connect, so accidental changes to critical routing are less likely.

4. Why the DL32SE replaced the old Yamaha mixer

The previous church mixer was a 32-channel Yamaha analog mixer. It was functional for traditional room sound, but the expanding livestream and AV requirements made a digital rack mixer more practical.

Need / problem	How the DL32SE helps
Separate livestream sound from room sound	Aux sends allow a dedicated stream mix that can be adjusted independently from the sanctuary main mix.
Remote control from the sound position or tablet	Master Fader SE allows control from a PC or tablet on the AV network. The operator does not have to stand at the rack.
Repeatable settings	Snapshots and saved shows make it easier to restore a known-good setup for Sunday service.
More flexible routing	Assignable outputs can feed sanctuary sound, auxiliary amps, overflow spaces, monitor sends, or a camera/audio path. USB routing remains available for future OBS use if configured.
Better integration with streaming	The current stream path uses Axis embedded audio received by OBS. USB/ASIO can be used later only after configuration changes.
Less physical rewiring for changes	Many routing changes can be made in software rather than moving cables or repatching an analog board.

Important practical reason

The DL32SE was not chosen simply because it is newer. It was chosen because streaming requires a dependable, repeatable, separate broadcast audio mix. The old analog mixer was less convenient for this kind of routing, recall, and remote operation.

5. How the mixer fits into the streaming system

The DL32SE is the central audio routing device for the church. Master Fader SE controls the DL32SE mixer. Stream Agent III sd controls and monitors OBS Studio; it does not carry or process the church audio. In the current setup, OBS receives embedded audio from the Axis camera source(s). USB audio from the DL32SE is not currently used for OBS, but it could be enabled later with OBS and Stream Agent configuration changes.

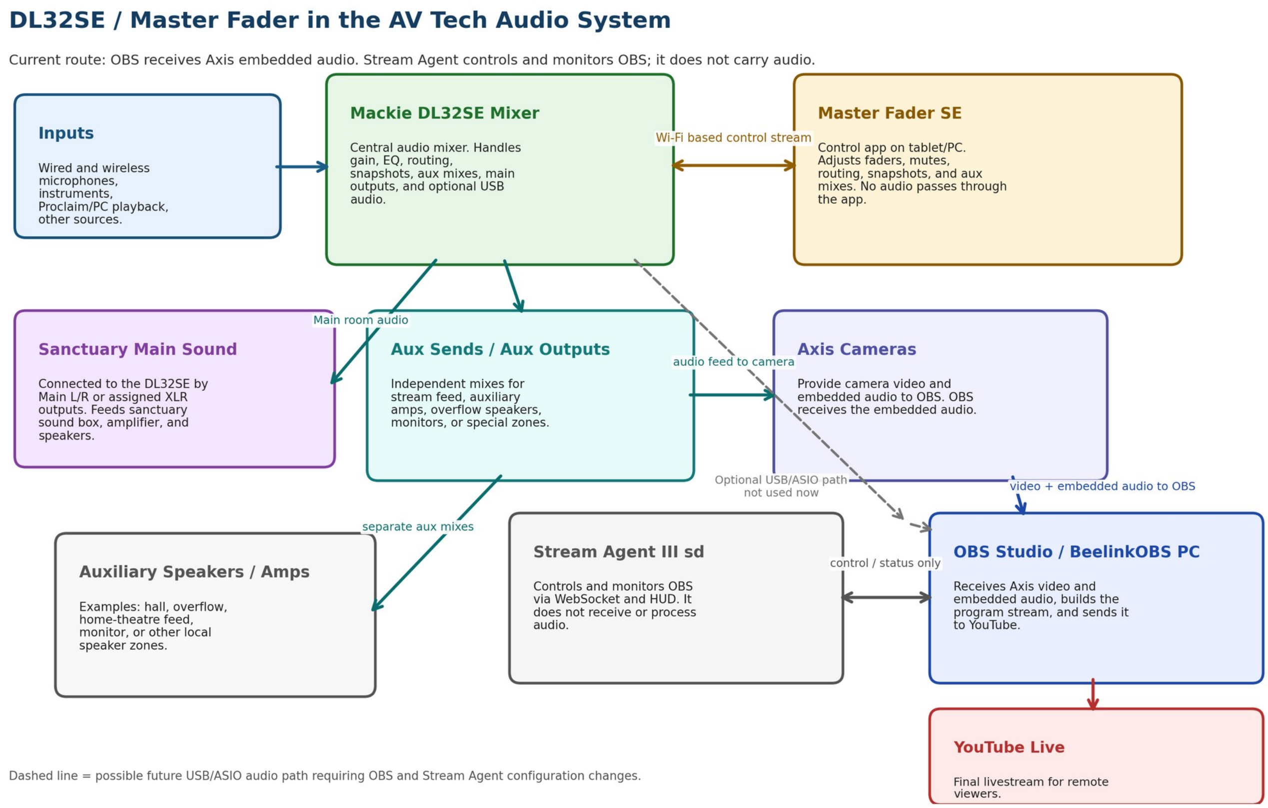

Figure 1 - Conceptual role of the DL32SE and Master Fader SE in the AV tech system.

6. Main room mix versus stream mix

A major reason for the DL32SE is that the mix heard in the room is not always the same as the mix needed online. The room already contains natural sound from voices, instruments, and speakers. Online listeners hear only what the stream receives.

Mix	Typical destination	Main concern
Main L/R or main assigned output	Sanctuary sound box, amplifier, and speakers	Comfortable sound level and clarity for people physically in the room.
Dedicated stream aux mix, often Aux 3 in this project history	Axis camera embedded audio into OBS Studio, then YouTube Live	Clear speech and music for online listeners. Stream Agent does not carry this audio.
Other aux outputs	Auxiliary amps, overflow speakers, monitors, hall/home-theatre feed, special zones	Give each area only the sound it needs, at its own level.

In the current setup, OBS receives the stream audio as embedded audio from the Axis camera source(s). The DL32SE/Master Fader system remains the audio-control point, but Stream Agent is not in the audio path. Aux 3 or another DL32SE output may still be used as a feed to the camera/audio system; the exact physical output and cable path should be verified in Master Fader SE and recorded in the future screenshot-based revision of this document.

7. Understanding aux sends and aux outputs

An aux send is an independent send level from each input channel to an auxiliary mix bus. An aux output is the physical or routed output that carries that aux mix to another device. The important idea is that each aux mix can be different from the main room mix.

- A channel can be loud in the room but quieter in the livestream aux mix.
- A channel can be muted or low in an auxiliary speaker zone if that source is not needed there.
- A stream aux can be set post-fader so normal service fader changes still follow the operator, while retaining separate stream-send trims.
- Monitor or auxiliary speaker feeds may be pre-fader or post-fader depending on whether they should follow main fader moves.
- A single physical aux output can feed an amplifier, powered speaker, camera input, interface input, or other destination if levels and cabling are correct.

7.1 Pre-fader versus post-fader concept

Aux tap type	Plain-English meaning	Typical use
Pre-fader	The aux send is less affected by the channel fader used for the room mix.	Stage monitor or special feed where the room fader should not disturb that mix.
Post-fader	The aux send follows the channel fader. If the operator turns the microphone up/down for the room, the aux feed tends to follow.	Livestream mix where the stream should generally follow the live service mix, but still have its own send levels.
Pre-DSP / other tap points	The send may be taken before some processing, depending on mixer settings.	Advanced cases only. Avoid changing unless the reason is documented.

Recommended operating idea

For the YouTube stream mix, a post-fader aux is often the easiest volunteer-friendly choice: the main fader move still helps the stream, but the stream aux send level can be adjusted separately for sources that need different online balance.

8. Suggested/documented audio routing model

The following model reflects the project history and current Stream Agent III sd design. It should be checked against the live Master Fader SE routing screen and corrected if the physical system has changed.

Route	Purpose	Notes / verify later
Inputs -> DL32SE channels	All microphones, instruments, and playback enter the mixer.	Label channels clearly in Master Fader SE.
DL32SE Main/assigned outputs -> sanctuary main sound box	Feeds the sanctuary sound box, amplifier, and speakers for the room.	This is the room-sound path. Do not tune it only from the YouTube return; tune the room independently.

Route	Purpose	Notes / verify later
DL32SE output / stream feed -> Axis camera audio embedding	Provides the audio that is carried with the Axis camera source(s).	Current system uses Axis embedded audio into OBS. Verify the exact DL32SE output, sound-box connection, and cable path.
Axis camera embedded audio -> OBS Studio	Current audio path for the YouTube stream.	OBS receives this audio. Stream Agent controls and monitors OBS but does not receive audio.
Other aux sends -> auxiliary amps/speakers	Feeds overflow/hall/home-theatre/monitor-style zones.	Each aux mix can be independent. Label cables and outputs.
PC/Proclaim playback -> mixer stereo return or input channels	Allows videos/music from Proclaim/PC to be heard.	Avoid routing playback back into itself; watch for feedback loops.
DL32SE USB/ASIO -> OBS Studio (optional, not current)	Alternative direct computer-audio path.	Not currently used. Requires OBS and Stream Agent configuration changes before use.

9. Operating overview for Sunday service

1. Power on the AV cabinet equipment as documented for the site.
2. Confirm the DL32SE is powered and connected to the AV network through Ethernet.
3. Confirm the Master Fader SE app launches on BeelinkOBS or the control tablet.
4. Confirm Master Fader SE can see/connect to the DL32SE.
5. Recall or verify the correct church show/snapshot if that workflow is used.
6. Check that the main room microphones and music sources are active and at reasonable levels.
7. Check the dedicated stream/feed mix and confirm OBS receives the Axis embedded audio.
8. Use Master Fader SE for audio mixing; use Stream Agent III sd for OBS stream workflow and director control.
9. After service, leave the system in the documented state for the next operator.

10. Master Fader SE areas to document later with screenshots

Screenshots are not included in this first draft. The following areas should be captured later and inserted into this document.

Screenshot area	Why it matters
Mixer connection screen	Shows how to connect Master Fader SE to the DL32SE on the AV network.
Main mixer/fader view	Shows channel layout and naming used by volunteers.
Aux 3 / Stream Mix view	Shows the dedicated stream mix levels and tap point/post-fader behavior.
Output routing screen	Shows which aux/output paths feed sanctuary sound, Axis camera audio embedding, auxiliary amps, or speakers.
USB routing / I/O patch screen	Needed only if the future USB/ASIO OBS path is enabled. It is not part of the current audio path.
Snapshots/shows page	Shows how to restore the normal church setup.
Access limiting / user permissions	Important if tablets or volunteer devices are allowed to connect.

11. Troubleshooting

Symptom	Likely causes	First checks
Master Fader cannot find mixer	Mixer not powered, Ethernet cable problem, wrong network, AP/router issue, Windows firewall/network profile.	Check DL32SE power/link lights, ER605/AP network, BeelinkOBS IP, and whether tablet/PC is on the AV LAN.
Room sound works but stream is silent	Axis camera audio not embedded or selected in OBS, camera/audio source disabled, stream feed to camera missing, or wrong OBS/Stream Agent audio-mode assumption. USB audio is not the current path.	Check OBS audio meter for the Axis camera source, confirm the Axis embedded-audio source is active, check camera/audio feed, and verify Stream Agent is controlling OBS rather than receiving audio.
Stream level too low	Stream feed level to the Axis camera too low, camera input/embedded-audio level low, OBS gain low, or mixer channels not feeding the stream/camera path.	Use Master Fader SE to check the stream/feed mix, then check Axis embedded-audio level and OBS meters for healthy speech peaks without clipping.
Stream level noisy when mixer off	Audio cables or camera/interface inputs may be floating or picking up induced noise.	Power mixer on for normal operation; check cable routing away from AC power; verify balanced connections where possible.
Tablet loses control of mixer	AP/network instability, weak Wi-Fi, mixer Ethernet problem, router/switch issue.	Check whether BeelinkOBS Master Fader also disconnects. If both disconnect, suspect wired/network gear before tablet Wi-Fi.
Auxiliary speaker too loud/quiet	Wrong aux mix, wrong output level, amplifier volume, or channel send levels.	Open the relevant Aux mix in Master Fader and confirm output routing and send levels.
Feedback or echo	A PC playback/return loop, open microphone near speakers, or wrong aux routing.	Mute suspect returns one at a time; ensure stream return/YouTube audio is not fed back into the mixer.

12. Maintenance notes

- Keep a known-good Master Fader SE show/snapshot backed up after major changes.
- Label physical outputs and cables in the AV cabinet: Main, Stream Aux, Aux Speaker, Camera/OBS feed, etc.
- Document the final OBS audio path: currently Axis embedded audio into OBS. USB/ASIO audio is optional and not active unless OBS and Stream Agent configuration are changed.
- Avoid making routing changes during a live service unless the current failure requires it.
- After any routing or firmware/app update, test room sound, stream audio, auxiliary speakers, and recording before Sunday service.

13. Relationship to Stream Agent III sd

Stream Agent III sd is not a replacement for Master Fader SE. Stream Agent controls and monitors OBS Studio, Web HUD controls, MIDI cue handling, timer/start/stop support, and related stream workflow items. Master Fader SE controls the audio mixer itself. In the current audio routing, OBS Studio receives the embedded audio from the Axis camera source(s); Stream Agent does not receive or process audio. Both systems are required for the complete church AV workflow.

System	Primary responsibility
Master Fader SE / DL32SE	Audio mixing, aux sends, output routing, snapshots, mixer processing, stream mix.
Stream Agent III sd	OBS control and monitoring, Web HUD, MIDI cue handling, timer/start/stop support, and Director preview. It does not receive or process audio.
OBS Studio	Video/audio encoding and YouTube streaming. In the current setup, OBS receives embedded audio from the Axis camera source(s).
ER605 / AV network	Connectivity between mixer, PC, cameras, tablet, remote access, and internet path.

14. Source list

The following sources were used for manufacturer facts and terminology. Site-specific decisions are based on the church AV project history and should be verified against the live system.

Source	Used for
Mackie DL32SE product page, mackie.com	DL32SE overview, Master Fader SE role, features, inputs/outputs, USB, aux sends.
Mackie DL16SE/DL32SE Spec Sheet, mackie.com	DL32SE specifications and feature list.
Mackie Master Fader product page, mackie.com	Master Fader / Master Fader SE app purpose, supported families, features, access limiting, downloads.
Mackie Master Fader v5.1 Reference Guide URL supplied by user	Detailed app reference, aux send/source concepts, and future setup verification.